

Exhibit 6.6 shows how this rule performed with the same 50/50 portfolio in the sixty-two rolling historical periods from US history since 1926 for a 4 percent initial withdrawal rate. Occasionally the popular press will mistakenly define the 4 percent rule this way (withdraw 4 percent of the remaining account balance each year), but the accepted definition of the 4 percent rule is the constant inflation-adjustment spending strategy defined earlier. To be clear, with constant inflation-adjusted spending, the withdrawal rate will change throughout retirement as the portfolio value changes. The withdrawal rate adjusts, while spending stays the same. But with the fixed-percentage rule, the withdrawal rate stays the same, while spending adjusts.

Spending is naturally more volatile with the fixed-percentage strategy. After thirty years, spending ranges from \$2.67 to \$9.10 in real terms, relative to the initial \$4.00 spent from a \$100.00 portfolio. At the same time, remaining wealth is less volatile than before, as the range after thirty years is \$76.71 to \$230.74. In both cases, spending and wealth were lower in the worst cases from the middle part of retirement, approaching low points of \$2.00 from a portfolio with \$50.00 left. In the historical data, poor market environments in early retirement (which led to these low points) were followed by better market environments in the latter part of retirement. Because spending declined along with the portfolio as the spending rate remained constant at 4 percent, portfolio losses were not locked in to the same extent, and the spending rate was not forced upward to meet the spending goal, thus avoiding sequence-of-returns risk, helping preserve the portfolio, and letting it subsequently grow. The disadvantage, as the exhibit clearly reveals, is that spending paths were quite volatile with this strategy throughout history.

Exhibit 6.6

Time Path of Real Spending and Wealth

Fixed-Percentage Withdrawals

For 4% Initial Spending Rate, 50/50 Asset Allocation, Rolling 30-Year Retirements

Using SBBI Data, 1926–2016, S&P 500 and Intermediate-Term Government Bonds